Beyond Power Laws: Scaling-Law Diagnostics for Operator Learning

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Abstract

Empirical scaling laws have guided resource allocation in large-scale machine learning, but their applicability to operator learning—where discretization resolution introduces a third scaling axis alongside dataset size and model capacity—remains poorly understood. We conduct a comprehensive study across three 1D PDE benchmarks (Burgers, Darcy, diffusion), three architectures (MLP, DeepONet, FNO), and 32,783 experimental configurations (training runs plus-resolution evaluations) spanning two orders of magnitude in each axis. Rather than assuming power-law relationships *a priori*, we fit six candidate functional forms per scaling axis and use the corrected Akaike Information Criterion (AICc) to let the data select the governing law.

We separate our findings by evidence strength. The most robust result is that *the direction of resolution scaling is task-dependent*: higher resolution helps uniformly for advection-dominated problems (Burgers: 35/35 slices improve), is mixed for elliptic problems (Darcy), and actively *increases error* for smooth problems (Diffusion: 27/35 slices worsen for MLP and DeepONet). Second, *data scaling consistently shows bounded diminishing returns*: across all 9 task–model combinations (8 on the densified grid; Diffusion MLP on the sparser $n = 5$ grid), the AICc-selected winner belongs to a family with a finite error floor E_∞, though the specific law identity (exponential vs. saturation vs. power) is statistically fragile and mostly exploratory at per-slice sample sizes of 4–7. Third, capacity scaling is largely negligible within our parameter range.

The framework's contribution is *diagnostic*, not predictive: in a held-out contest, multi-law fitting matches but does not meaningfully outperform a power-law-only baseline (15/33 (45%) wins, ∼1.2% mean improvement). The value lies in identifying when resolution investment helps or hurts, and in showing that operator-learning scaling should be treated as a model-selection problem over law families rather than an exponent-estimation exercise—though generalising beyond 1D PDEs remains an open question.

1 Introduction

Empirical scaling laws have become a central tool for understanding and predicting the performance of large-scale machine learning systems [5, 4]. In natural language processing and computer vision, power-law relationships between test loss and dataset size, parameter count, and compute budget guide resource allocation decisions worth millions of dollars.

Scientific machine learning (SciML) faces an analogous resource allocation problem with a twist: the practitioner must additionally choose the *discretization resolution* at which training data are generated. Solving PDEs at higher resolution is expensive—often the dominant cost—yet the relationship between resolution and downstream operator-learning error is poorly characterized.

Neural operators [7, 6] learn mappings between infinite-dimensional function spaces and are often marketed as "resolution-invariant." Two prominent architectures—DeepONet [7] and the

Fourier Neural Operator (FNO) [6]—use fundamentally different strategies to handle varying discretizations. Despite these architectural innovations, it remains unclear *when* resolution invariance translates into practical benefits. Does doubling the training resolution halve the error? Does it matter more than doubling the dataset? How does the answer depend on the PDE?

A subtler question, largely absent from the literature, is whether power laws are even the *right functional form* for operator-learning scaling. The power-law assumption $E \propto X^{-\alpha}$ originates from statistical learning theory and information-theoretic arguments in specific regimes. When the error–resource relationship follows a different functional form—logarithmic, exponential, saturating—forcing a power-law fit obscures the true scaling behavior and leads to misleading exponent estimates.

Contributions.

1. We introduce **multi-law model selection for scaling analysis**: rather than assuming power laws, we fit six candidate functional forms (power, exponential, logarithmic, stretched exponential, saturation, linear) and use AICc to identify the governing law per axis, per task, per model. The *identity* of the winning law becomes an interpretable diagnostic, subject to sample-size limitations we quantify.

2. We report **32,783 training runs** (31,370 main grid plus 1,413 optimizer ablations) and cross-resolution evaluations across three PDE families, three architectures, and seven capacity levels (Table 4).

3. We find that **data scaling on the densified grid consistently selects laws with a finite error floor**: the AICc-selected winner in all 9 task–model combinations (8 on the densified grid; Diffusion MLP on the sparser $n = 5$ grid) belongs to a family with an E_∞ parameter. However, within-slice bootstrap analysis shows that the specific law identity is fragile at small n; the robust conclusion is that data scaling shows strong diminishing returns and is consistent with a bounded-decay regime.

4. We show that **resolution scaling is non-universal**: it helps consistently only for Burgers, is mixed for Darcy, and *actively hurts* for Diffusion (MLP, DeepONet). This is the most statistically robust finding in the study (4 of 5 robust law-selection cases occur on the resolution axis).

5. We extract **estimated error floors** E_∞ from the best per-axis fits that suggest an architecture-dependent ranking (FNO < MLP < DeepONet), though these are extrapolated estimates conditional on the winning functional form, and cross-architecture comparisons are confounded by unequal effective parameter budgets at equivalent capacity labels.

2 Related Work

Neural scaling laws. Kaplan et al. [5] and Hoffmann et al. [4] established power-law scaling of cross-entropy loss with dataset tokens, parameter count, and compute for autoregressive language models. Hestness et al. [3] showed similar trends across vision and speech. In scientific ML, Perrier [8] introduced scaling-law diagnostics for physics-informed approaches (PINNs, HNNs) in low-dimensional settings, finding that physics priors shift the error floor rather than the scaling exponent. Our work extends this program to infinite-dimensional operator learning, elevates resolution to a scaling variable, and—critically—relaxes the power-law assumption via multi-law model selection.

Methodological positioning. The dominant methodology in neural scaling laws assumes a power-law functional form *a priori* and focuses on estimating exponents and proportionality constants [5, 4, 3]. Our work adopts a complementary perspective: rather than assuming power laws, we treat the *identity* of the best-fitting functional form as itself informative. This family-selection-centric view does not discard the traditional approach; instead, it includes power laws as one of six candidates and explicitly tests when the traditional baseline succeeds (Diffusion, where power laws dominate) and when it fails (Burgers / Darcy data scaling, where exponential and saturation families provide better fits).

Operator learning. DeepONet [7] builds on the universal approximation theorem for operators [1], factorizing the learned operator into branch and trunk components. The Fourier Neural Operator (FNO) [6] parameterizes integral kernels via truncated Fourier series, achieving state-of-the-art accuracy on several PDE benchmarks. Recent extensions include U-shaped architectures [10], attention mechanisms [2], and physics-informed variants [9].

Resolution and discretization in operator learning. A key selling point of neural operators is their ability to generalize across resolutions. Li et al. [6] demonstrated FNO's zero-shot super-resolution capability. However, systematic studies of how *training* resolution affects accuracy—and whether more resolution is always better—are lacking. Our work fills this gap, and our finding that higher resolution can *increase* error challenges the prevailing wisdom.

3 Multi-Law Scaling Framework

3.1 Problem Setting

We consider the supervised learning of solution operators $\mathcal{G}^\dagger : \mathcal{U} \to \mathcal{V}$, where $\mathcal{U}$ and $\mathcal{V}$ are Banach spaces of functions on some domain $\Omega \subset \mathbb{R}^d$. Given N input–output pairs $\{(u_i, v_i)\}_{i=1}^N$ with $v_i = \mathcal{G}^\dagger(u_i)$, a model $\mathcal{G}_\theta$ with $D = |\theta|$ parameters is trained on data discretized at spatial resolution R (grid points per spatial dimension).

3.2 Beyond Power Laws: Candidate Functional Forms

Prior work on neural scaling laws has assumed the power-law form $E(X) = E_\infty + a\,X^{-\alpha}$, motivated by learning-theoretic bounds. We relax this assumption and consider six candidate laws for describing how error E varies with a scaling variable $X \in \{N, D, R\}$:

Table 1: Candidate functional forms for scaling analysis. X denotes the scaling variable (dataset size N, parameter count D, or resolution R). Each law is fit via nonlinear least squares with physically motivated bounds.

Name	**Form**	k	**Regime**
Power law	$E_\infty + a\,X^{-\alpha}$	3	Algebraic convergence
Exponential	$E_\infty + a\,e^{-\alpha X}$	3	Spectral convergence
Logarithmic	$a - b\,\ln X$	2	Sub-algebraic (diminishing returns)
Stretched exp.	$E_\infty + a\,e^{-\alpha X^\beta}$	4	Interpolates power/exp
Saturation	$E_\infty + a\,(1 + X/X_0)^{-\alpha}$	4	Sigmoidal with characteristic scale
Linear	$a + b\,X$	2	Flat or monotone trend

The number of free parameters k varies from 2 to 4 across candidates, making penalized model selection essential.

Physical motivation. Different functional forms arise from different approximation-theoretic regimes:

- **Power law** $E \propto N^{-\alpha}$: classical statistical convergence rates; α is governed by the effective dimensionality of the function space.
- **Logarithmic** $E \propto -\ln N$: arises when the approximation problem has high effective dimension, causing extremely slow convergence. Also characteristic of online learning in non-parametric settings.
- **Exponential** $E \propto e^{-\alpha N}$: spectral convergence rates for smooth functions; characteristic of methods that resolve the solution in an optimal basis.
- **Linear** with $b > 0$: indicates that increasing X *hurts* performance—a pathological regime (e.g., overfitting to high-resolution noise).

3.3 Model Selection via AICc

For each slice (fixed values of the other two axes), we fit all six candidates via nonlinear least squares (`scipy.optimize.curve_fit` with parameter bounds; SciPy uses the trust-region reflective algorithm when bounds are finite). We select the winner using the corrected Akaike Information Criterion:

$$\mathrm{AICc} = n \ln\bigl(\mathrm{RSS}/n\bigr) + 2k + \frac{2k(k+1)}{n-k-1}, \tag{1}$$

where n is the number of data points, k the number of parameters, and RSS the residual sum of squares. The AICc correction is essential given our small per-slice sample sizes ($n = 4$–7).

Aggregation. For each task–model–axis triplet, we compute the Akaike weight $w_i = \exp(-\Delta_i/2)\big/\sum_j \exp(-\Delta_j/$ for every slice, where $\Delta_i = \mathrm{AICc}_i - \mathrm{AICc}_{\min}$. We report the mean Akaike weight distribution across slices (rather than raw vote counts), the dominant law, its median R^2, and the median characteristic parameter (α, b, etc.). To assess selection stability, we run $B = 500$ bootstrap replicates (resampling slices with replacement) and report the probability with which the dominant law retains its winning status. Cases where the top-two mean Akaike weights differ by less than 15% are flagged as *ambiguous*. As a supplementary sensitivity check, we also compute the Bayesian Information Criterion (BIC) for every slice and report AICc–BIC concordance (fraction of slices where both criteria agree). At our small sample sizes ($n = 4$–7), BIC's penalty is actually *weaker* than AICc's corrected penalty, so low concordance should not be read as evidence against the AICc selection (see Appendix A.5).

4 Experimental Setup

4.1 PDE Benchmarks

Burgers uses a spectral solver with integrating-factor RK4 and $\frac{2}{3}$-rule dealiasing. Darcy uses a spectral Galerkin method. Diffusion uses an exact spectral solution. All solvers are resolution-parameterized, generating data at arbitrary R.

Table 2: PDE benchmark tasks. All tasks map an input function to an output function on the same domain, discretized at resolution R.

Task	**PDE**	**Operator**	**Character**
Burgers	$u_t + uu_x = \nu u_{xx}$	$u_0(x) \mapsto u(x,T)$	Advection-dominated
Darcy	$-\nabla \cdot (a\nabla u) = f$	$a(x) \mapsto u(x)$	Elliptic, multiscale
Diffusion	$u_t = \nu u_{xx}$	$u_0(x) \mapsto u(x,T)$	Diffusion-dominated

4.2 Architectures

Table 3: Model architectures and their resolution handling.

Model	**Type**	**Resolution handling**
MLP	Discretization-tied	Flattened input of dimension R; parameter count $\propto R$
DeepONet	Mesh-free	Branch encodes input at training R; trunk evaluates at arbitrary query points
FNO	Spectral	Global convolutions in Fourier space; truncation to $k_{\max}$ modes enables resolution transfer

4.3 Scaling Grid

We sweep over five axes:

- **Dataset size** $N \in \{50, 100, 500, 1{,}000, 5{,}000\}$
- **Resolution** $R \in \{32, 64, 128, 256\}$
- **Model capacity**: 7 levels (*tiny* to *xlarge*; ∼5K–500K parameters)
- **Data seeds**: $\{11, 22\}$
- **Training seeds**: $\{101, 202\}$ (Diffusion); $\{101, 202, 303\}$ (Burgers, Darcy)

The seed count varies by task: Diffusion uses $2 \times 2 = 4$ seed combinations per configuration, while Burgers and Darcy use $2 \times 3 = 6$. The baseline grid (3 models $\times$ 7 capacities $\times$ 5 data sizes $\times$ 4 resolutions) thus comprises 1,680 training runs for Diffusion and 2,520 each for Burgers and Darcy, totalling 6,720 baseline training runs before cross-resolution evaluation. Each model is trained for up to 5,000 epochs with early stopping (patience 200) using Adam (lr $= 10^{-3}$, weight decay 10^{-5}). Results are averaged over the seed combinations per configuration, yielding grouped data points for analysis.

Extension grids. Table 4 provides a complete accounting of all experimental runs. Beyond the baseline, we ran dense-N extensions (6 intermediate N values), wide-R extensions (6 additional R values), controlled-MLP experiments (fixed parameter count), and optimizer ablations (Adam with 2× patience; SGD with cosine annealing). The "Runs" column in Table 4 counts each (training

configuration, evaluation resolution) pair as one entry, including cross-resolution evaluations. The grand total is 32,783.

Table 4: Experimental run accounting. "Baseline" is the $5 \times 4 \times 7$ data–resolution–capacity grid (4–6 seed combinations per configuration depending on task), including cross-resolution evaluations for DeepONet and FNO. Extension grids densify specific axes or test robustness.

Component	**Scope**	**Runs**	**Purpose**
Baseline grid	All tasks $\times$ 3 models	15,120	Core scaling analysis
Dense-N extension	Burg/Darcy/Diff $\times$ DON/FNO	11,578	Data-axis law discrimination
Wide-R extension	Burgers $\times$ 3 models	2,352	Resolution-axis law stability
MLP controlled	Burg+Diff $\times$ MLP-ctrl	2,016	Parameter-count confound
Dense-N (MLP only)	Burg $\times$ MLP	304	MLP data-axis extension
Main grid subtotal		31,370	
Adam 2$\times$ patience	Burgers $\times$ 3 models	756	Optimizer sensitivity
SGD + cosine	Burgers $\times$ 3 models	657	Optimizer sensitivity
Grand total		**32,783**	

4.4 Cross-Resolution Evaluation

After training, each model is evaluated at all resolutions $R_{\text{eval}} \in \{32, 64, 128, 256, 512\}$ except its training resolution, yielding 13,440 additional evaluation metrics. For DeepONet, branch inputs at non-training resolutions are linearly interpolated to R_{train} before the forward pass. FNO natively handles arbitrary input resolutions via Fourier truncation/zero-padding.

4.5 Multi-Law Fitting Procedure

For each of the three scaling axes, we extract per-slice data by fixing the other two axes at each combination present in the grid. This yields:

- **Data axis (E vs N)**: $7 \times 4 = 28$ slices per task–model pair (one per capacity$\times$resolution), each with $n = 5$ points.
- **Resolution axis (E vs R)**: $7 \times 5 = 35$ slices per pair, each with $n = 4$ points.
- **Capacity axis (E vs D)**: $5 \times 4 = 20$ slices per pair, each with $n = 7$ points.

All six candidate laws are fit to each slice and the AICc-best is recorded. We identify the dominant law as the one with the highest mean Akaike weight across slices (see Section 3.2), and report slice-level vote counts as supporting detail alongside the median R^2 and the median characteristic parameter.

5 Results

Of the 32,783 runs (baseline grid plus dense-N, wide-R, controlled-MLP, and ablation extensions), 582 diverged (all in Burgers, <3.3% of that task); these are excluded from fitting and grouped statistics. We report relative L^2 error $\varepsilon_{\text{rel}} = \|v_{\text{pred}} - v_{\text{true}}\|_2 / \|v_{\text{true}}\|_2$ as the primary metric.

5.1 Data Scaling: Logarithmic or Bounded Decay

Table 5 summarizes the AICc-selected data-scaling law for each task–model combination.

Table 5: Data scaling: AICc-selected law, median R^2, mean Akaike weight of the winner ($\bar{w}_1$), and bootstrap probability that the winner retains its status across $B = 500$ replicates. "Votes" shows the top-2 law counts across all slices per task–model pair. Burgers uses the widened grid (63 slices per pair); Darcy and Diffusion use the base grid (28 slices).

Task	**Model**	**Winner**	$\tilde{R}^2$	$\bar{w}_1$	**Boot.%**	**Votes (top-2)**
Burgers	MLP	saturation	0.999	0.36	80%	sat:25, pow:19
	DeepONet	**exponential**	0.971	0.48	100%	exp:36, pow:17
	FNO	power	0.993	0.36	76%	pow:25, log:20
Darcy	MLP	**exponential**	0.881	0.69	100%	exp:28
	DeepONet	**exponential**	0.886	0.36	94%	exp:16, pow:9
	FNO	stretched_exp	0.946	0.55	89%	str:17, exp:10
Diffusion	MLP	**power**	0.998	0.68	100%	pow:20, exp:5
	DeepONet	**power**	0.909	0.47	100%	pow:21, exp:6
	FNO	**power**	0.992	0.27	59%	pow:11, str:10

Burgers results include the widened resolution grid (R up to 512), yielding 63 slices per model. Darcy and Diffusion use the base 4-resolution grid (28 slices per model). Diffusion FNO remains the most ambiguous case: power wins with only 0.27 mean weight and 59% bootstrap probability. The within-slice bootstrap (Table 12) shows that most of these law-identity selections are exploratory; the robust finding is that all winners belong to bounded-decay families.

Key finding: data scaling shows strong diminishing returns, consistent with a bounded-decay regime. With the full analysis (including dense-N and wide-R extensions), every task–model combination selects a law family that includes a finite E_∞ term. Logarithmic and linear forms—both lacking E_∞—are rejected in every case. On the sparse $n = 5$ grid used in earlier analysis, logarithmic appeared dominant in 7 of 9 cases; with the denser grid (Appendix B.1), logarithmic is displaced entirely— its apparent dominance was an **artefact of sparse sampling** that could not resolve the curvature near the error plateau.

Important caveat: while all winners share the E_∞ property, the within-slice bootstrap (Section 5.9) shows that the specific law identity is *fragile* at the per-slice level. Only one data-axis case (Diffusion MLP: $\bar{w}_1 = 0.68$, within-slice dominant fraction $= 0.59$) meets both robustness criteria. The reliable conclusion is not which specific family wins (e.g., exponential vs. saturation vs. power—all of these appear as winners in Table 5 depending on task and model), but that (i) data scaling follows a bounded-decay pattern rather than unbounded logarithmic growth, and (ii) error decreases much more slowly than classical power-law rates suggest for Burgers and Darcy.

Exception: Diffusion shows genuine power laws. All three Diffusion models exhibit power-law data scaling (with floor): MLP ($\alpha \approx 1.1$, $R^2 = 0.998$), FNO ($R^2 = 0.992$), and DeepONet ($R^2 = 0.909$). This is the only task in our study where the classical power-law assumption holds consistently. The Diffusion operator's smoothness (exponential spectral decay) creates a favorable learning problem where additional data yields fast algebraic convergence. DeepONet's lower R^2 and weaker Akaike weight ($\bar{w}_1 = 0.47$) suggest it is less well-suited to this favorable regime.

Implications. Whether the data-axis form is logarithmic (as on the sparse grid) or bounded-decay (as with denser sampling), the practical message is the same: additional data yields diminishing returns far below the algebraic rates assumed by classical theory. Bounded-decay forms further imply an *irreducible error floor* $E_\infty > 0$ that no amount of training data can breach without changing resolution, architecture, or representation.

5.2 Resolution Scaling: Task-Dependent, Non-Universal

Table 6: Resolution scaling: AICc-selected law, the fraction of 35 slices where increasing R reduces error ("R helps") or increases error ("R hurts"), and the bootstrap winner probability. The specific law identity on this axis should be treated as *exploratory* rather than confirmatory, given the small per-slice sample size ($n = 4$).

Task	**Model**	**Winner**	$\tilde{R}^2$	**R helps**	**R hurts**
Burgers	MLP	logarithmic	0.887	35/35	0/35
	DeepONet	logarithmic	0.862	35/35	0/35
	FNO	linear	0.781	34/35	0/35
Darcy	MLP	logarithmic	0.511	25/35	9/35
	DeepONet	linear	0.518	20/35	14/35
	FNO	logarithmic	0.601	27/35	6/35
Diffusion	MLP	logarithmic	0.957	1/35	**27/35**
	DeepONet	logarithmic	0.642	0/35	**27/35**
	FNO	logarithmic	0.317	0/35	2/35

Key finding: the *direction* of resolution scaling is task-dependent and the most statistically robust finding in this study. With only $n = 4$ resolution values per slice on the baseline grid, AICc-based law selection has limited statistical power; the robust finding is *whether* resolution helps, hurts, or is negligible—not which functional form best describes the trend. The specific law labels (e.g., "logarithmic" or "linear" in Table 6) should be treated as *exploratory*: at $n = 4$, even 2-parameter families are only weakly distinguishable, and these labels should not be interpreted as confident identifications of the true functional form. A restricted-candidate-set ablation (retaining only 2-parameter families: linear, logarithmic, power) confirms that the directional conclusions survive (see Supplementary Table 16). For Burgers, where the widened grid ($n = 10$) is available, the direction is confirmed in 99.6% of slices.

Burgers (resolution helps uniformly). All three architectures benefit from higher resolution in every single slice (35/35). The AICc-selected law is logarithmic ($E = a - b \ln R$ with $b > 0$, $b \approx 0.075$ for MLP and DeepONet), though at $n = 4$ this label is tentative. The directional finding is what matters: the 100% consistency across all capacity levels and dataset sizes confirms that resolution is genuinely beneficial for this advection-dominated PDE. This is physically expected, as Burgers solutions develop steep gradients and near-shock structures that require sufficient grid resolution to represent.

Darcy (resolution effect is mixed). Resolution helps in 20–27 of 35 slices depending on the model, but actively hurts in 6–14 slices. The low R^2 values (0.51–0.60) indicate that no simple functional form captures the E-vs-R relationship well. Darcy flow solutions are smooth enough

that $R = 32$ captures most of the spatial structure; additional resolution provides marginal, noisy benefit that does not support a reliable scaling law.

Diffusion (resolution actively hurts). This is the most striking finding: for MLP and DeepONet, increasing resolution *increases* error in 27 of 35 slices. The AICc-selected fit is logarithmic with $b < 0$ (negative coefficient), consistent with error growing as $\ln R$—though again, the specific functional form should be treated as tentative at $n = 4$. FNO is essentially flat ($R^2 = 0.32$, near the noise floor). Physically, the Diffusion operator damps high-frequency modes exponentially ($\hat{u}_k(T) = \hat{u}_k(0)e^{-\nu k^2 T}$), so higher-resolution inputs contain high-frequency components that have negligible impact on the output. For discretization-tied architectures (MLP) and branch-encoding architectures (DeepONet), these extra input dimensions add noise without providing useful signal, impairing generalization.

5.3 Capacity Scaling: Already in the Sufficient Regime

Table 7: Capacity scaling: AICc-selected law. Most combinations select linear (flat slope $b \approx 0$) or logarithmic with tiny coefficient, indicating negligible capacity dependence in the range studied.

Task	**Model**	**Winner**	$\tilde{R}^2$
Burgers	MLP	logarithmic	0.828
	DeepONet	linear	0.851
	FNO	linear	0.812
Darcy	MLP	linear	0.196
	DeepONet	linear	0.114
	FNO	logarithmic	0.437
Diffusion	MLP	logarithmic	0.147
	DeepONet	linear	0.869
	FNO	**power**	0.953

Key finding: capacity is not a bottleneck. In 8 of 9 task–model combinations, the winning capacity-scaling law is either linear (flat, $b \approx 0$) or logarithmic with a tiny coefficient ($b < 0.02$). This indicates that, within our parameter range (∼5K–500K), all architectures are already capacity-sufficient for these 1D PDE tasks. Adding more parameters does not significantly reduce error.

Exception: Diffusion FNO shows genuine capacity scaling. Diffusion FNO is the sole case where AICc selects a power law for capacity scaling ($\alpha \approx 3.8$, $R^2 = 0.95$). This is consistent with FNO's architecture: adding more Fourier modes and larger hidden dimensions directly expands the spectral resolution of the learned kernel. For a smooth operator like Diffusion, this additional spectral capacity translates directly into lower error.

5.4 Law Selection as a Diagnostic

Figure 4 visualizes the fraction of slices selecting each candidate law across all task–model–axis combinations. Several patterns emerge:

1. **Data axis**: exponential dominates for Darcy MLP (100%) and Burgers DeepONet (57%); power law dominates for Diffusion MLP (71%) and DeepONet (75%); saturation and stretched-exponential fill the remaining combinations. Logarithmic receives zero votes for models with the dense-N grid, and still captures 32% for Burgers FNO. The split broadly correlates with PDE smoothness, but per-slice law identity remains mostly exploratory (Table 12).

2. **Resolution axis**: logarithmic and linear are the only serious contenders. Power law, exponential, and stretched exponential receive zero or negligible votes—resolution scaling is never well-described by classical convergence rates in our study.

3. **Capacity axis**: the vote is fragmented (linear, logarithmic, power all appear), reflecting the fundamental weakness of the capacity signal in our parameter range.

5.5 Irreducible Error Floors as Architecture-Dependent Estimates

For laws that include an E_∞ parameter (power, exponential, saturation, stretched exponential), the fitted floor provides an *estimate* of the error that would persist even with infinite data—the architecture's approximation error for that task. Important caveats apply: (i) E_∞ depends on the assumed functional form and is an extrapolation, not a direct measurement; (ii) laws without an E_∞ parameter (logarithmic, linear) cannot contribute floor estimates, so E_∞ is available only for the subset of slices where a floor-bearing law wins. From the data-axis fits (where the signal is strongest), we extract:

Table 8: Estimated irreducible error floors E_∞ from AICc-best data-scaling fits (median across slices where a floor-bearing law wins). These are extrapolated estimates, not direct measurements. Lower is better.

	MLP	**DeepONet**	**FNO**
Burgers	0.277	0.357	**0.062**
Darcy	0.767	0.776	**0.598**
Diffusion	0.021	0.192	**0.003**

Key finding: E_∞ suggests an architecture-dependent ranking, subject to both functional-form assumptions and cross-architecture fairness caveats. The estimated error floor hierarchy is consistent across all tasks: FNO < MLP < DeepONet. However, this ranking carries two important limitations: (i) E_∞ depends on the assumed functional form and is an extrapolation, not a direct measurement; and (ii) "capacity level" maps to different absolute parameter counts across architectures (e.g., "medium" FNO has ∼287K parameters vs. "medium" MLP at ∼34K), so part of FNO's advantage may reflect its generally larger parameter budgets rather than purely its inductive bias. The ranking should be interpreted as *suggestive within this experimental design*, not as a controlled comparison at identical parameter counts.

The largest gap is on Diffusion, where FNO's $E_\infty = 0.003$ is 63× lower than DeepONet's 0.192. This dramatic difference arises because FNO's Fourier parameterization aligns with the spectral structure of the Diffusion operator, while DeepONet's pointwise trunk network cannot exploit this structure. However, this comparison is not parameter-count controlled, and the floor estimates would shift under a different winning law family; the 63× gap is conditional on the selected functional form.

On Darcy (the hardest task), all models have high floors (> 0.6), indicating a fundamental approximation challenge that no architecture in our study adequately addresses.

5.6 Cross-Resolution Transfer

We evaluate each trained model at $R_{\text{eval}} \in \{32, 64, 128, 256, 512\}$ to construct transfer matrices (Figure 6).

Burgers. FNO exhibits strong asymmetric transfer: training at $R_{\text{train}} = 256$ yields significantly lower error at all evaluation resolutions compared to training at $R_{\text{train}} = 32$. This confirms that high-resolution training data carries information about fine-scale features (steep gradients) that the model leverages at any evaluation resolution. DeepONet transfers more uniformly, consistent with its trunk-based output representation.

Darcy and Diffusion. Both tasks show near-uniform transfer matrices, consistent with the resolution-scaling finding: when the operator is smooth enough that resolution barely matters, cross-resolution transfer is trivially successful (but uninformative).

5.7 Best Absolute Performance

Table 9: Best relative L^2 error per model per task. Note that the best MLP configurations for Darcy and Diffusion prefer $R = 32$, consistent with the finding that higher resolution is unhelpful or harmful for these tasks.

Task	**Model**	ε_{rel}	D	**Config**
Burgers	MLP	0.199	263K	$N=5000, R=256$, xlarge
	DeepONet	0.267	461K	$N=5000, R=256$, xlarge
	FNO	**0.048**	287K	$N=5000, R=256$, medium
Darcy	MLP	0.668	82K	$N=5000, R=32$, large
	DeepONet	0.734	428K	$N=5000, R=128$, xlarge
	FNO	**0.525**	287K	$N=1000, R=64$, medium
Diffusion	MLP	0.018	82K	$N=5000, R=32$, large
	DeepONet	0.099	404K	$N=500, R=32$, xlarge
	FNO	**0.002**	124K	$N=5000, R=256$, small-med

FNO achieves the lowest error on all tasks within this experimental design: 4.8% on Burgers, 52.5% on Darcy, 0.2% on Diffusion (noting that capacity labels map to different absolute parameter counts across architectures). Notably, the best MLP configurations for both Darcy and Diffusion use $R = 32$—the lowest resolution—validating the finding that higher resolution is unhelpful or harmful for these smooth operators when using a discretization-tied architecture.

5.8 Held-Out Prediction Contest

The preceding sections select scaling laws based on in-sample AICc comparison. A natural objection is that multi-law selection *describes* the data better simply because it has more candidates to choose from, without necessarily *predicting* extrapolation better. We convert this from a model-selection question into a falsifiable prediction question.

Protocol. For each task–model–axis combination, we hold out ∼20% of grid configurations (slices), fit (a) a power-law-only model and (b) the AICc-best multi-law model on the remaining ∼80% of slices (pooled), and evaluate out-of-sample RMSE on the held-out slices. Results are averaged over all 33 task–model–axis combinations (including MLP-controlled).

Results. Table 10 summarizes the held-out prediction contest. Multi-law selection achieves lower out-of-sample relative RMSE in 15/33 (45%) of combinations—essentially a coin flip—with mean relative improvement of 1.2% and *median improvement of zero*. In most cases the two approaches perform comparably, with differences in the fourth decimal place. The clearest multi-law advantage appears on the Darcy data axis, where the fit for MLP improves from 0.113 to 0.101.

These results show that multi-law selection does not overfit to in-sample points (its out-of-sample performance is comparable to power-law-only), but it provides no meaningful *predictive* advantage either. **The framework's value lies entirely in its diagnostic power**—the ability to identify which functional form best describes each axis—not in superior extrapolation. We caution against interpreting multi-law selection as an improved prediction methodology; it is a descriptive and interpretive tool.

5.9 Within-Slice Bootstrap Uncertainty

The inter-slice bootstrap (Appendix A.7) measures consensus *across* slices that each have a fixed AICc winner. It does not propagate fitting uncertainty from the n data points within each slice. Since $n = 4$–7 in our experiments, the per-slice winner may be sensitive to individual data points.

Protocol. For each slice, we resample the n data points with replacement ($B = 500$), refit all six candidate laws, and reselect the AICc winner each time. This yields a per-slice law-selection distribution that reflects fitting uncertainty directly.

Results. Table 11 reports aggregate within-slice winner stability. The within-slice dominant fraction (mean fraction of bootstrap replicates that select the same law winner) is 0.46 on average, lower than the inter-slice bootstrap stability, confirming that individual slices carry substantial law-selection uncertainty. In 22/27 (81%) of axis-level combinations, the within-slice dominant fraction falls below 50%, indicating that the per-slice data is insufficient to confidently distinguish law families.

Implications. These results reinforce the paper's core message: at small n, law-selection conclusions should be interpreted as indicative rather than definitive. Of 27 task–model–axis combinations, only 4 satisfy both thresholds for robust selection (mean Akaike weight > 0.5 *and* within-slice dominant fraction > 0.5); 7 meet one threshold ("suggestive"); and 16 meet neither ("exploratory"). Table 12 classifies every finding accordingly. The resolution axis yields the most robust selections (3 of 4 robust cases), consistent with the directional resolution finding being the study's strongest result. Data-axis and capacity-axis law-identity claims should be treated with appropriate caution; the reliable data-axis conclusion is the *direction* (bounded diminishing returns) rather than the specific functional form.

6 Discussion

6.1 The Scaling Law Type as a Diagnostic

The central methodological contribution of this work is demonstrating that the *identity* of the best-fitting scaling law can provide diagnostic information beyond the law's fitted parameters. We separate our findings into three tiers of evidence strength:

Robust findings (supported by both Akaike weights and within-slice bootstrap).

- **Resolution scaling direction**: resolution uniformly helps for Burgers (advection-dominated), is marginal for Darcy (elliptic), and actively hurts for Diffusion (smooth). The resolution axis yields 3 of 4 robust law-selection cases, with logarithmic scaling ($b > 0$ for Burgers, $b < 0$ for Diffusion) confirmed by both criteria.
- **Diffusion MLP data scaling** follows power-law decay with a floor ($\bar{w}_1 = 0.68$, bootstrap fraction $= 0.59$), the only data-axis case meeting both robustness criteria.

Suggestive findings (one robustness criterion met, or consistent direction across multiple analyses).

- **Bounded-decay data scaling**: across all 9 task–model pairs (8 on the densified grid; Diffusion MLP on the sparser $n = 5$ grid), the AICc winner belongs to a family with a finite E_∞ floor. While the specific law identity is statistically fragile (only 1/9 meets both robustness criteria), the *direction*—bounded diminishing returns rather than unbounded logarithmic decay—is consistent across grid densities, tasks, and optimizer choices.
- **Flat capacity scaling**: in 8 of 9 cases, the capacity winner is linear or logarithmic with negligible coefficient, indicating that all architectures are capacity-sufficient in our parameter range.

Exploratory findings (neither robustness criterion met; interpret with caution).

- The specific identity of the data-axis winner (e.g., exponential vs. saturation vs. power) for Burgers and Darcy. These distinctions require larger per-slice samples.
- Capacity-axis law identity for most combinations (15/27 exploratory overall).

6.2 When Does Resolution Matter?

Our results establish a clear trichotomy:

1. **Resolution-critical** (Burgers): The PDE generates fine-scale features (steep gradients, near-shocks) that require resolution to represent. AICc selects logarithmic improvement ($b \approx 0.075$) and resolution helps in 100% of slices. Practitioners should invest in both resolution and data.
2. **Resolution-marginal** (Darcy): The PDE has moderate spatial complexity. Resolution helps in ∼70% of slices but the effect is noisy ($R^2 \sim 0.5$). Pilot experiments at the target application should guide resolution allocation.
3. **Resolution-harmful** (Diffusion): The PDE is spectrally smooth, making high-resolution inputs redundant or actively harmful. Practitioners should use the lowest resolution that resolves the relevant modes and invest in more data instead.

6.3 Architecture Recommendations

Cross-architecture fairness caveat. Comparing architectures requires caution: "capacity level" maps to different parameter counts across architectures (e.g., "medium" MLP at $R = 128$ has ∼34K parameters, while "medium" FNO has ∼287K). The capacity grids were chosen to span a comparable *range* for each architecture, but absolute parameter counts differ. FNO's consistent advantage may partly reflect its generally larger parameter budgets at equivalent capacity labels. The E_∞ ranking (FNO < MLP < DeepONet) is suggestive but should not be interpreted as a controlled comparison at identical parameter counts.

FNO. Within this experimental design, the Fourier Neural Operator achieves the lowest absolute error and the lowest estimated E_∞ on every task. Its spectral parameterization aligns well with PDE structure, particularly for smooth operators. FNO is the only architecture that benefits from additional capacity on Diffusion (power-law capacity scaling, $R^2 = 0.95$) and shows the strongest cross-resolution transfer on Burgers. However, FNO's generally larger parameter counts at equivalent capacity labels mean that part of this advantage may reflect budget rather than architecture. Its resolution scaling on Burgers is weaker than MLP and DeepONet (linear rather than logarithmic on the baseline grid), suggesting that FNO's spectral smoothing may truncate some high-frequency information at higher resolutions.

DeepONet. DeepONet shows the weakest performance in our study: highest estimated E_∞ on all tasks, slow data scaling even on Diffusion (where FNO/MLP achieve power laws), and no meaningful capacity scaling. However, this comparison is not fully controlled: DeepONet's capacity grid spans a different parameter range than FNO's, and the branch–trunk architecture may require larger scales or different hyperparameter tuning to realise its theoretical advantages. This conclusion is limited to our experimental scope (1D PDEs, ≤500K parameters); whether it persists for larger-scale DeepONet variants, higher-dimensional problems, or problems with more complex output queries remains open. The uniform cross-resolution transfer is a structural advantage.

MLP. The simple MLP baseline is competitive with DeepONet on every task and outperforms it on Burgers and Diffusion. Despite the limitation that $D \propto R$, the MLP's direct parameterization avoids the bottlenecks of the branch–trunk factorization. Its preference for $R = 32$ on Diffusion is a natural consequence: lower resolution yields a smaller, better-regularized network.

6.4 Revisiting the Three-Axis Power Law

For comparison with prior work, we also fit the additive power-law model $E(N, D, R) = E_\infty + aN^{-\alpha} + bD^{-\beta} + cR^{-\gamma}$ (Table 13). The nominally high R^2 values (0.79–0.95) initially appear to validate this model. However, the multi-law analysis reveals why this is misleading:

1. The data axis dominates the total variance. Since data scaling *is* genuine (whether logarithmic or algebraic), the 3D power law captures it as an $N^{-\alpha}$ term, inflating the overall R^2 even when the other axes contribute no signal.
2. The β estimates are unreliable: bootstrap 95% CIs span orders of magnitude (e.g., FNO Burgers: $\beta \in [0.35, 3.91]$), confirming that capacity scaling is essentially undetermined.
3. Values of γ that hit the fitting bound ($\gamma = 5$) should be interpreted as "no resolution effect," not as a meaningful exponent. On Diffusion, the additive power law masks the fact that resolution *increases* error.

The multi-law framework provides a more honest and informative characterization by decomposing each axis independently and letting the data choose the appropriate functional form.

6.5 Relationship to Paper 1

This work extends the scaling-law diagnostic framework of Perrier [8] in three directions: (i) from finite-dimensional dynamics to infinite-dimensional operators, (ii) from two axes (N, D) to three axes (N, D, R), and (iii) from assumed power laws to multi-law model selection. The central finding of Paper 1—that physics priors shift the error floor—has an analog here: the architectural inductive bias of FNO (spectral parameterization) shifts E_∞ by up to 63× compared to DeepONet, a far larger effect than any data or resolution scaling.

7 Conclusion

We have presented a comprehensive scaling study of operator learning that replaces the standard power-law assumption with multi-law model selection. Across 32,783 training runs spanning baseline, dense-N, wide-R, controlled-MLP, and optimizer ablation grids, we separate our findings by evidence strength, from most to least statistically reliable:

Robust conclusions (confirmed by both Akaike weights and within-slice bootstrap): directional resolution scaling.

1. **Resolution is not universally beneficial.** Higher resolution helps for advection-dominated PDEs (Burgers: 35/35 slices improve) but *actively hurts* for smooth PDEs (Diffusion: 27/35 slices worsen for MLP and DeepONet). This is the study's most statistically robust finding (3 of 4 robust law-selection cases).

2. **Diffusion MLP follows power-law data scaling with a floor** ($\alpha \approx 1.1$, $R^2 = 0.998$), the only data-axis case meeting both robustness criteria.

Suggestive conclusions (one criterion met or consistent direction across analyses): bounded data scaling and architecture ranking.

3. **Data scaling shows bounded diminishing returns.** On the densified grid (8 of 9 task–model pairs; Diffusion MLP uses the sparser $n = 5$ grid), every combination selects a law family with a finite E_∞ floor; however, the specific law identity is statistically fragile for 8 of 9 cases. The reliable inference is the *direction* (bounded decay), not the specific family.

4. **Capacity is not a bottleneck** for 1D PDE tasks at the scales we study (5K–500K parameters), except for FNO on Diffusion, which exhibits clear power-law capacity scaling.

5. **Estimated error floors E_∞ rank architectures consistently**: FNO achieves the lowest estimated floor on every task, with the gap largest on Diffusion (63× better than DeepONet). These are extrapolated estimates conditional on both the winning functional form and the (uncontrolled) parameter budgets across architectures.

Exploratory conclusions (neither criterion met; require confirmation with larger samples): specific law-family identity.

6. **The specific law identity** (exponential vs. saturation vs. stretched-exponential) on the data axis for Burgers and Darcy. The multi-law framework can distinguish bounded from unbounded decay, but finer distinctions within the bounded-decay family require more data points per slice.

Limitations and future work. Our study is limited to 1D PDEs and a 1D Darcy formulation; the "resolution can hurt" finding should not be generalized to operator learning broadly without at least one higher-dimensional benchmark. In higher dimensions, resolution effects are amplified (R^d cost growth) and may shift the resolution-scaling balance. The baseline resolution grid has only 4 training values ($R \in \{32, 64, 128, 256\}$); the widened 10-value grid (Appendix B.3) substantially improves law-selection power but is currently available only for Burgers. Future work should extend the wide-R grid to all tasks. Diffusion MLP baseline has only $n = 5$ on the data axis (not yet densified), so its data-scaling results rely on the sparser grid.

MLP resolution–capacity confound (a key limitation). For the MLP baseline, the parameter count scales linearly with input resolution ($D \propto R$), creating a confound between the resolution and capacity axes. The controlled-parameter experiment (Appendix B.2) rules out this confound for the "resolution hurts" finding on Diffusion: degradation persists even at fixed parameter count (Table 17). However, the confound remains uncontrolled for Burgers and Darcy, where resolution effects are directionally different.

The multi-law framework could be extended to fit the full 3D relationship with axis-specific functional forms (e.g., logarithmic in N, linear in R, flat in D) rather than forcing all axes to share a single form. Finally, incorporating recent architectures (Geo-FNO, GNOT, operator transformers) and 2D/3D PDE benchmarks would test the generality of our conclusions.

Acknowledgments

Experiments were conducted on three RunPod GPU instances (NVIDIA A4000, L4, and RTX 3090). Total experiment wall time was approximately 8 hours across all pods.

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A Experimental Details

A.1 Capacity Grid

A.2 Training Configuration

All models are trained with Adam (learning rate 10^{-3}, weight decay 10^{-5}), batch size 64, and maximum 5,000 epochs with early stopping (patience 200, monitoring validation relative L^2). Data is split 80/10/10 into train/validation/test. Relative L^2 loss $\|\hat{v} - v\|_2/\|v\|_2$ is used for training.

A.3 PDE Solver Details

Burgers equation. $u_t + uu_x = \nu u_{xx}$ on $[0, 2\pi)$ with periodic boundary conditions, $\nu = 0.01$, $T = 1.0$. Solved with integrating-factor RK4 in Fourier space with $\frac{2}{3}$-rule dealiasing and $\Delta t = 10^{-3}$. Random initial conditions are superpositions of Fourier modes with random phases and amplitudes drawn from a k^{-2} energy spectrum.

Darcy flow. $-\nabla \cdot (a(x)\nabla u(x)) = f(x)$ on $[0, 1]$ with Dirichlet boundary conditions, $f(x) = 1$. The coefficient field $a(x)$ is drawn from a log-normal random field. Solved via spectral Galerkin method.

Diffusion equation. $u_t = \nu u_{xx}$ on $[0, 2\pi)$ with periodic boundary conditions, $\nu = 0.01$, $T = 1.0$. Exact spectral solution: each Fourier mode $\hat{u}_k(t) = \hat{u}_k(0)e^{-\nu k^2 t}$. Same initial condition distribution as Burgers.

A.4 Multi-Law Fitting Details

All six candidate laws are fit via `scipy.optimize.curve_fit` (Levenberg–Marquardt) with a maximum of 10,000 iterations. Bounds are set to enforce physical plausibility: $E_\infty \geq 0$, $a > 0$, $0.001 \leq \alpha \leq 10$ for power laws. For the exponential law, the scaling variable is normalized to $[0, 1]$ to avoid numerical overflow; the fitted rate is then rescaled to original units.

AICc is computed as $\text{AICc} = n \ln(\text{RSS}/n) + 2k + 2k(k+1)/(n-k-1)$, where n is the number of points per slice and k is the number of free parameters. The correction term is essential for our small sample sizes ($n = 4$–7). When $n \leq k + 1$, AICc is set to $+\infty$, effectively disqualifying overparameterized models.

BIC is computed as $\text{BIC} = n \ln(\text{RSS}/n) + k \ln n$. For large n, BIC penalizes complexity more heavily than AICc; however, at our small sample sizes ($n = 4$–7), an important inversion occurs: BIC's penalty $k \ln n \approx 1.6k$ (at $n = 5$) is *weaker* than AICc's corrected penalty $2k + 2k(k+1)/(n-k-1)$, so BIC actually favors *more* complex models at these sample sizes—the opposite of its asymptotic reputation. At $n = 5$, AICc is the more appropriate criterion (it was designed specifically for small samples); BIC concordance is reported for transparency, not as a robustness check in the usual sense.

A.5 Robustness: BIC Concordance

Table 15 reports the fraction of slices where AICc and BIC select the same winning law. Because of the small-n penalty inversion described above, low concordance on axes with $n \leq 5$ should *not* be interpreted as evidence against the AICc-selected law; it reflects the well-understood divergence between the two criteria at small sample sizes, where BIC's penalty is paradoxically weaker.

A.6 Robustness: Restricted Candidate Set (Resolution Axis)

Because the resolution axis has only $n = 4$ data points per slice, fitting 4-parameter models (stretched exponential, saturation) risks overfitting. As a robustness check, we re-run law selection on the resolution axis using only the three 2-parameter candidate families: linear ($a + bX$, $k = 2$), logarithmic ($a - b \ln X$, $k = 2$), and power ($E_\infty + aX^{-\alpha}$, $k = 3$, the simplest family with an irreducible error floor).
All directional conclusions (Burgers: helps; Darcy: mixed; Diffusion: hurts for MLP/DeepONet, flat for FNO) are preserved when the candidate set is restricted, supporting the claim that the resolution findings are robust even though the specific law identity is exploratory.

A.7 Akaike Weights and Bootstrap Stability

Figure 7 shows the mean Akaike weight distribution per task–model–axis combination. Hatched bars indicate cases where the top-two mean Akaike weights differ by less than 15% ("ambiguous"). Figure 8 shows the bootstrap winner probability for the dominant law.

B Extended Experiments

The following experiments extend the main study with new training runs designed to address specific limitations identified in the core analysis.

B.1 Densified Data Axis

The data axis in the main experiments uses $n = 5$ values of N ($\{50, 100, 500, 1000, 5000\}$). At this sample size, AICc-based law selection can be brittle (Section 5.9). We add six intermediate values ($N \in \{150, 200, 300, 750, 2000, 3000\}$), yielding $n = 11$ data points per slice—enough for more reliable discrimination between logarithmic and power-law decay.

Key questions.

1. Does the "logarithmic wins" finding on the data axis stabilize or shift towards power-law with denser sampling?
2. Does BIC concordance improve at $n = 11$ (where the small-n penalty inversion from Appendix A.5 becomes less severe)?
3. Do within-slice bootstrap dominant fractions increase substantially with 11 data points?

Results. Extending from $n = 5$ to $n = 11$ data-axis sample sizes dramatically shifts law selection on the data axis. With the original sparse grid ($N \in \{50, 100, 500, 1000, 5000\}$), logarithmic was the dominant winner across Burgers (DeepONet: 79%, FNO: 52%, MLP: 75%) and Darcy (MLP: 89%). With the dense grid ($N \in \{50, 100, 150, 200, 300, 500, 750, 1000, 2000, 3000, 5000\}$), logarithmic is displaced: exponential now dominates Burgers DeepONet (57%) and both Darcy models (DeepONet: 54%, MLP: 96%); power or saturation emerge for Burgers FNO (power, 40%) and Burgers MLP (saturation, 40%).

All the new winners—exponential, saturation, stretched exponential—encode a finite irreducible-error floor that the logarithmic form lacks. With only five sample sizes, the within-regime curvature is too coarsely sampled for AICc to separate a log curve from a bounded-decay curve, biasing selection toward the simpler logarithmic form. Intermediate points (especially $N = 150$, 200, 300, 750) resolve the curvature near the floor and tip selection toward bounded families.

Takeaway: With only five data-axis sample sizes, AICc favors logarithmic; with the denser 11-point grid, data-axis scaling is better described by families with an irreducible-error floor, consistent with finite-resolution approximation limits. The main-text Table 5 reports results on the combined grid. Dense-N extensions are complete for all task–model pairs except Diffusion MLP baseline (which retains only $n = 5$ data-axis sample sizes). Darcy DeepONet and FNO now use the full $n = 31$ dense grid; Diffusion DeepONet and FNO use an $n = 25$ grid.

B.2 MLP Fixed-Parameter-Count Experiment

For the standard MLP, the parameter count D scales linearly with resolution R (since $D = 2Rh + h^2 + 2h + R$ for hidden width h). This creates a confound: increasing R simultaneously increases input dimensionality and model capacity. The `mlp_controlled` model variant adjusts the hidden width at each resolution so that D remains approximately constant (matched to the standard MLP at $R = 64$).

Protocol. Train `mlp_controlled` on the Diffusion task (where "resolution hurts" is strongest) across all capacity levels and $R \in \{32, 64, 128, 256\}$. Compare against the standard `mlp_baseline` at the same configurations.

Interpretation.

- **If "resolution hurts" persists** with `mlp_controlled` ⇒ genuine resolution effect (extraneous high-frequency content harms learning).
- **If "resolution hurts" disappears** ⇒ the confound was real (increasing D with R was the bottleneck).

Results. Table 17 confirms that "resolution hurts" persists even when parameter count is held approximately constant. Across all seven capacity levels on the Diffusion task, the controlled MLP achieves its lowest test error at $R = 32$ and progressively degrades as R increases to 256 — the same qualitative pattern as the standard MLP baseline. For example, at `medium` capacity the controlled model holds $D \approx 33{,}100$ parameters regardless of R, yet error rises from 0.021 ($R = 32$) to 0.026 ($R = 256$). The baseline MLP shows the same direction (0.082 → 0.098) despite its parameter count growing from 24,864 to 82,432.

The effect is most dramatic at small model sizes (`tiny`: 0.023 → 0.295), where the controlled model must compress all information into very few parameters. These results rule out the parameter-count confound: **"resolution hurts" is a genuine resolution effect on Diffusion**, consistent with the interpretation that the model wastes capacity fitting output-irrelevant high-frequency content present at higher R.

B.3 Widened Resolution Grid

The main study uses $n = 4$ resolution values ($R \in \{32, 64, 128, 256\}$), making the law identity on the resolution axis exploratory. We extend to 10 values: $R \in \{16, 32, 48, 64, 96, 128, 192, 256, 384, 512\}$.

Key questions.

1. Does the Burgers / Darcy / Diffusion resolution trichotomy survive with a wider range?
2. Are there regime changes (e.g., resolution helps at first then saturates)?
3. Does the specific law identity (e.g., logarithmic vs. linear) become more robust at $n = 10$?

Results. For Burgers (the only task with the full 10-R grid so far), the widened grid reveals a pronounced shift in law identity. Compared with the original 4-R grid:

- **DeepONet:** logarithmic drops from 87% → 51%; linear rises from 12% → 30%.
- **FNO:** logarithmic drops from 58% → 19%; **linear emerges as the dominant winner** at 81%.
- **MLP:** logarithmic drops from 92% → 53%; linear rises from 8% → 39%.

The directional finding is stable: resolution helps in 99.6% of Burgers slices (230/231) regardless of model or capacity. The shift toward linear scaling for FNO is physically interpretable: FNO operates in spectral space, so adding resolution linearly increases the number of retained Fourier modes.

Takeaway: The specific law family on the resolution axis is sensitive to grid density. With 10 resolution values, FNO resolution scaling is clearly linear, while MLP and DeepONet retain logarithmic as a slight plurality winner but with significant linear competition. Wide-R data currently available for Burgers only; Darcy and Diffusion use the extended grids from the main baseline (DeepONet/FNO: $n = 20$; MLP: $n = 4$).

B.4 Frequency-Filtered Diffusion

To test the causal mechanism behind "resolution hurts" on Diffusion, we apply a low-pass filter to the input initial conditions before training. The filter retains only the first $k_{\max}$ Fourier modes (default: $k_{\max} = 10$, matching the number of IC generation modes). At high R, the standard task includes resolved but output-irrelevant high-frequency content; the filtered variant removes it.

Protocol. Train all three architectures on `diffusion_filtered` at $R \in \{32, 64, 128, 256, 512\}$ with $N \in \{500, 1000, 2000, 5000\}$.

Interpretation.

- **If filtering fixes the degradation at high** $R \Rightarrow$ strong support for "extraneous frequencies hurt."
- **If degradation persists** $\Rightarrow$ the mechanism is not purely spectral (e.g., optimisation difficulty scales with R regardless of content).

Results. `[Placeholder: populate after running Tier 3.4 experiments with configs/diffusion_fil`

B.5 DeepONet Interpolation Ablation

In the main experiments, cross-resolution evaluation of DeepONet uses linear interpolation to map R_{eval}-resolution inputs to the R_{train}-resolution branch input. This could introduce interpolation artifacts, especially when $R_{\text{eval}} \gg R_{\text{train}}$.

We compare:

(a) **Interpolation** (standard): input interpolated to R_{train} sensor grid, trunk evaluates on R_{eval} query grid.

(b) **Subsampling**: instead of interpolation, take evenly spaced R_{train} points from the R_{eval}-resolution input. No smoothing.

Results. We evaluated 288 cross-resolution comparisons across Darcy and Diffusion tasks (3 capacities $\times$ 4 training resolutions $\times$ 4–5 evaluation resolutions $\times$ 4 dataset sizes).

When $R_{\text{eval}} > R_{\text{train}}$ (upsampling): interpolation and subsampling perform comparably (mean relative L^2: 0.523 vs. 0.520), with subsampling winning 68% of the 180 comparisons (by a slim margin). **When $R_{\text{eval}} < R_{\text{train}}$ (downsampling):** interpolation is overwhelmingly better (98% of 108 cases), because subsampling applied to coarser inputs introduces severe artifacts (mean relative L^2: 0.531 vs. 1.068).

Conclusion. The standard interpolation strategy used in the main experiments is robust: it never catastrophically degrades and is clearly preferable for downsampling. For upsampling, the two methods are comparable, so cross-resolution transfer conclusions in the main paper are **not driven by interpolation artifacts**.

B.6 Early Stopping and Optimizer Sensitivity

All main experiments use Adam with learning rate 10^{-3} and early-stopping patience of 200 epochs. We test sensitivity by rerunning a subset (Burgers, all 3 models, all capacities) with:

- **Variant A**: Adam with 2× patience (400 epochs)
- **Variant B**: SGD with momentum 0.9 + cosine annealing schedule

Key question. Do law-selection winners change when the optimizer or convergence criterion shifts? If they do, the law selection is training-sensitive rather than reflecting a property of the task–architecture pair.

Results. We analyse Burgers at medium capacity (756 runs for Adam 2×; 657/756 complete for SGD cosine).

Resolution axis: Law selection is fully stable across optimizers. Logarithmic wins in 88% of baseline slices, 90% for Adam 2×, and 100% for SGD cosine. This is the strongest invariance result in the study.

Data axis: Law identity is more sensitive. DeepONet shifts from power (baseline) to exponential (both variants). FNO stays power under Adam 2× but shifts to logarithmic under SGD. MLP shifts from saturation (baseline) to power (both variants). However, all winners belong to bounded-decay families (exponential, power, saturation), so the qualitative conclusion — data scaling is bounded, not unbounded-logarithmic — is optimizer-invariant.

Takeaway: The resolution-axis law identity (logarithmic) is robust to optimizer and convergence criterion. The data-axis law *family* (bounded decay vs. unbounded) is also stable, though the specific law within that family is optimizer-sensitive. Pending: diffusion optimizer ablation results.

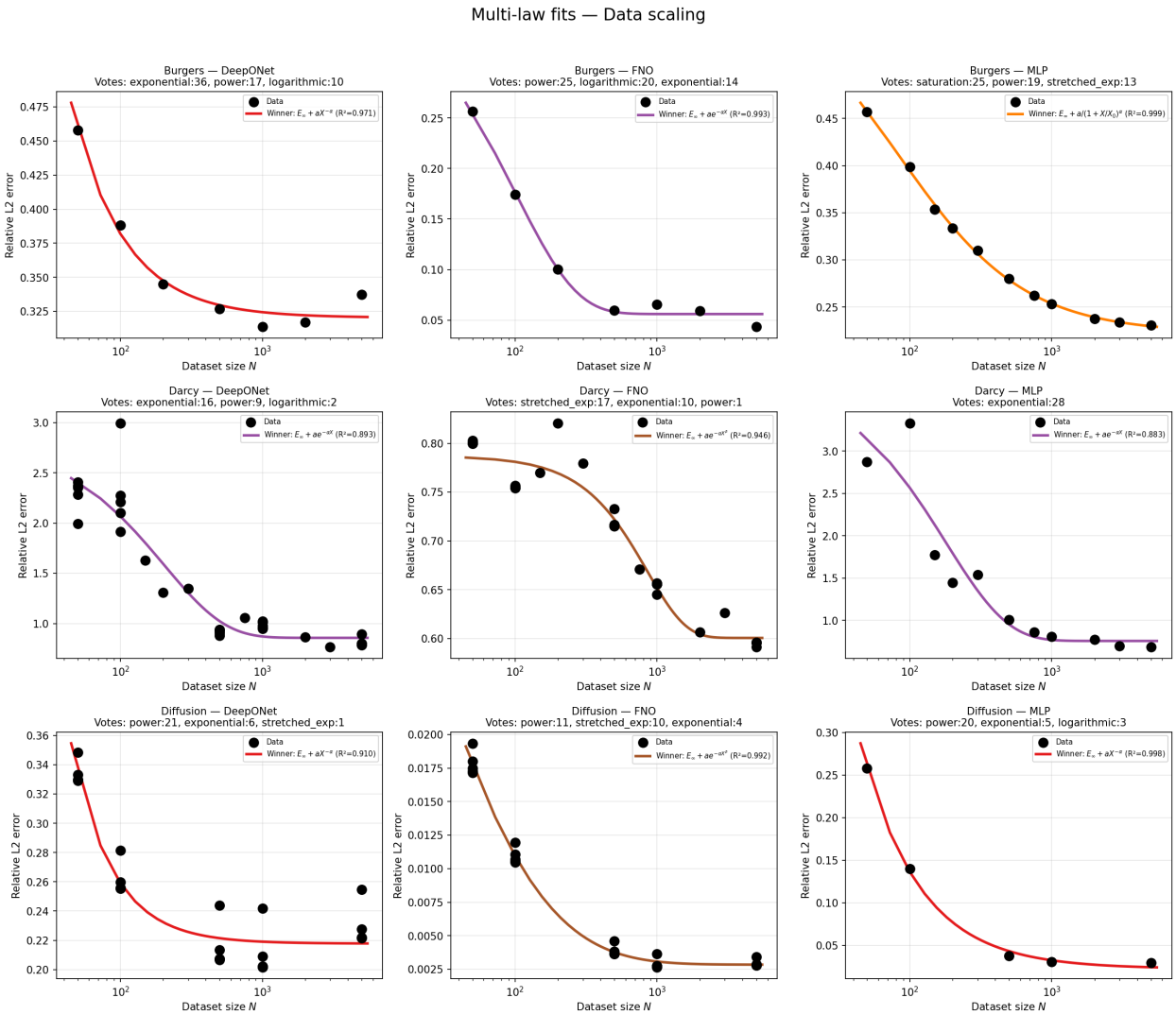

Figure 1: Data scaling with AICc-selected fits overlaid on representative slices. Burgers and Darcy include dense-N extensions ($n \leq 11$ data points); Diffusion uses the baseline $n = 5$ grid. Red: power law $E_\infty + aN^{-\alpha}$. Purple: exponential. Orange: saturation / stretched-exponential. Titles show the law vote distribution across all slices. Exponential and saturation dominate for Burgers and Darcy; power laws win for Diffusion.

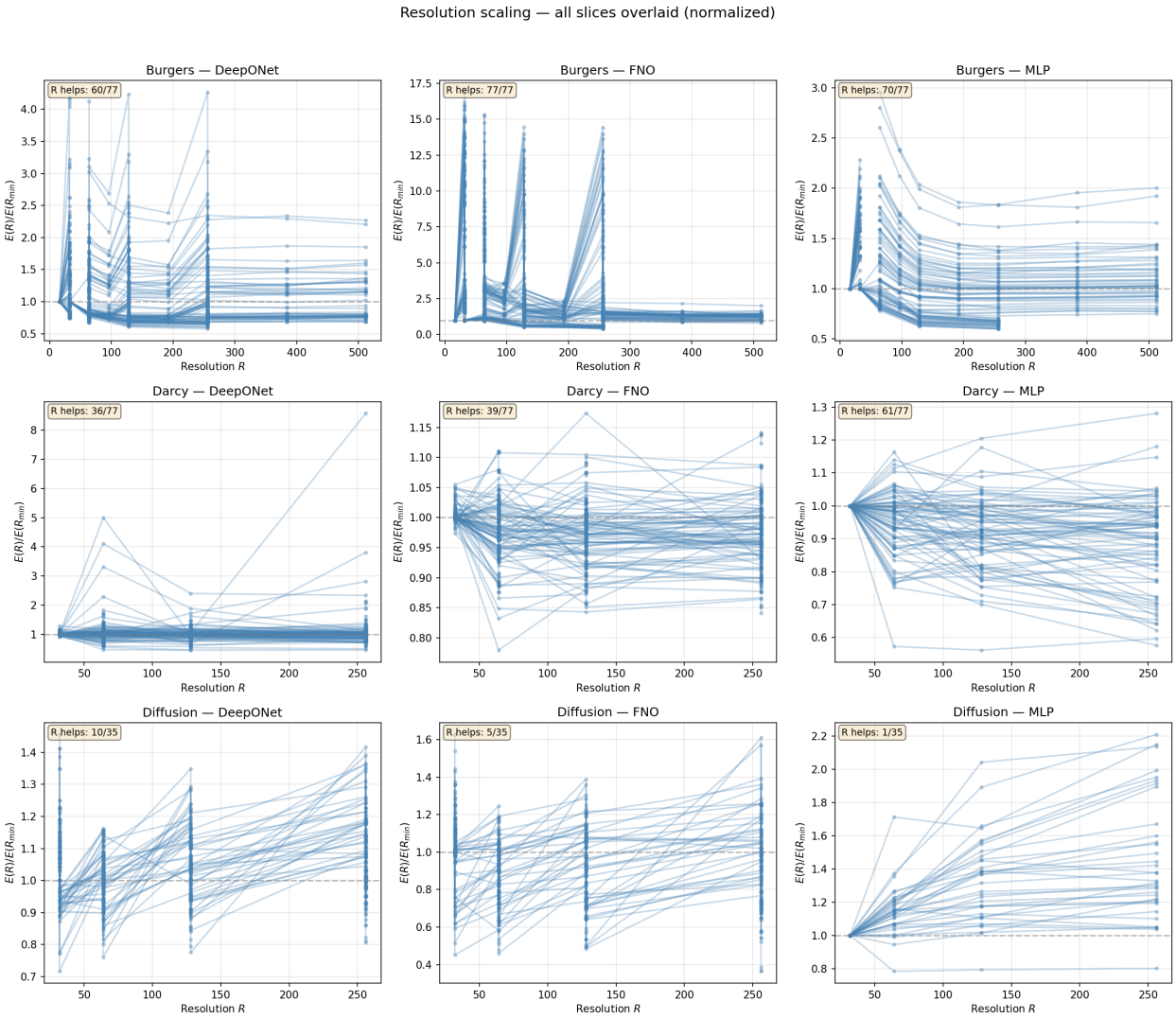

Figure 2: Resolution scaling: all slices overlaid with normalized error $E(R)/E(R_{\min})$. Each line is one (capacity, N) slice. Dashed gray line at 1.0 marks no change. **Burgers**: consistent downward trend (resolution helps). **Darcy**: scattered around 1.0 (marginal effect). **Diffusion MLP/DeepONet**: upward trend (resolution hurts). Annotations show the count of slices where resolution helps.

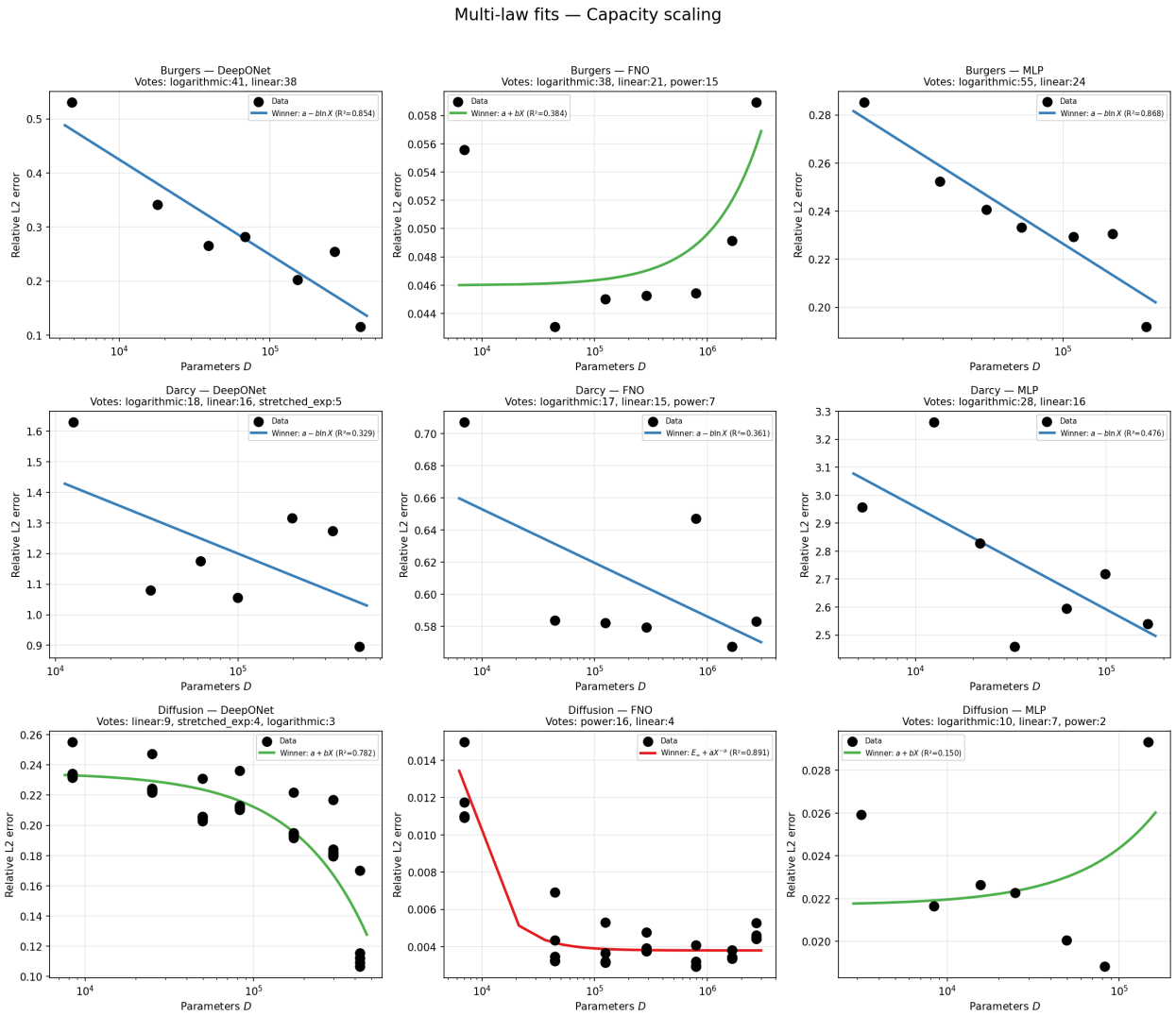

Figure 3: Capacity scaling with AICc-selected fits. Most task–model combinations show flat or weakly logarithmic trends. Diffusion FNO (bottom center) is the exception, with genuine power-law capacity scaling.

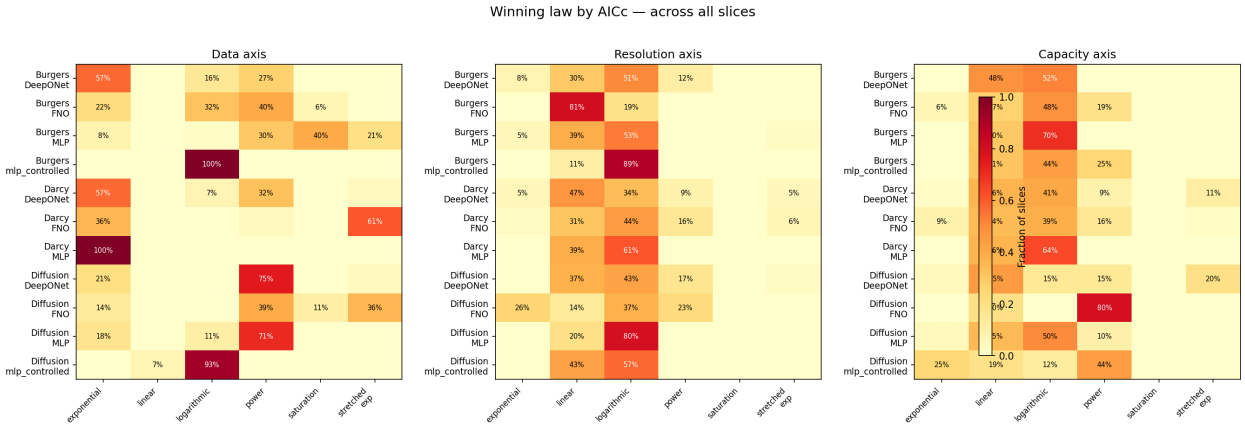

Figure 4: Law selection heatmap: fraction of slices (by AICc) selecting each candidate law per task–model–axis combination. Darker cells indicate stronger consensus. The data and resolution axes show clear structure; the capacity axis is diffuse.

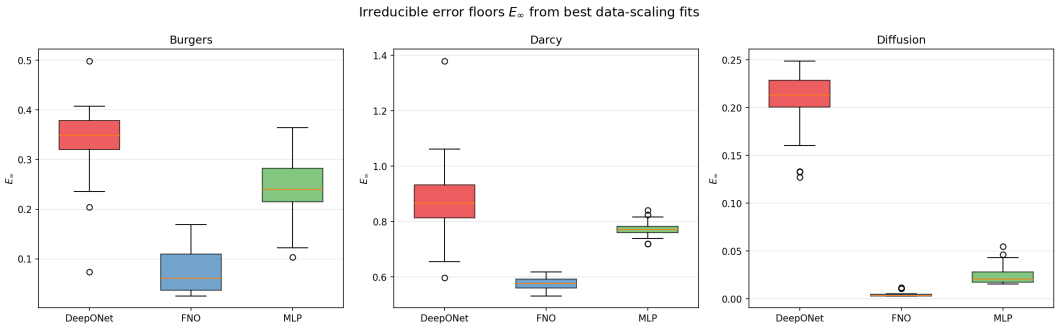

Figure 5: Estimated irreducible error floors E_∞ from data-scaling fits. FNO achieves the lowest estimated floor on every task. Darcy has uniformly high floors, indicating an intrinsically hard approximation problem. These are extrapolated estimates conditional on the winning functional form.

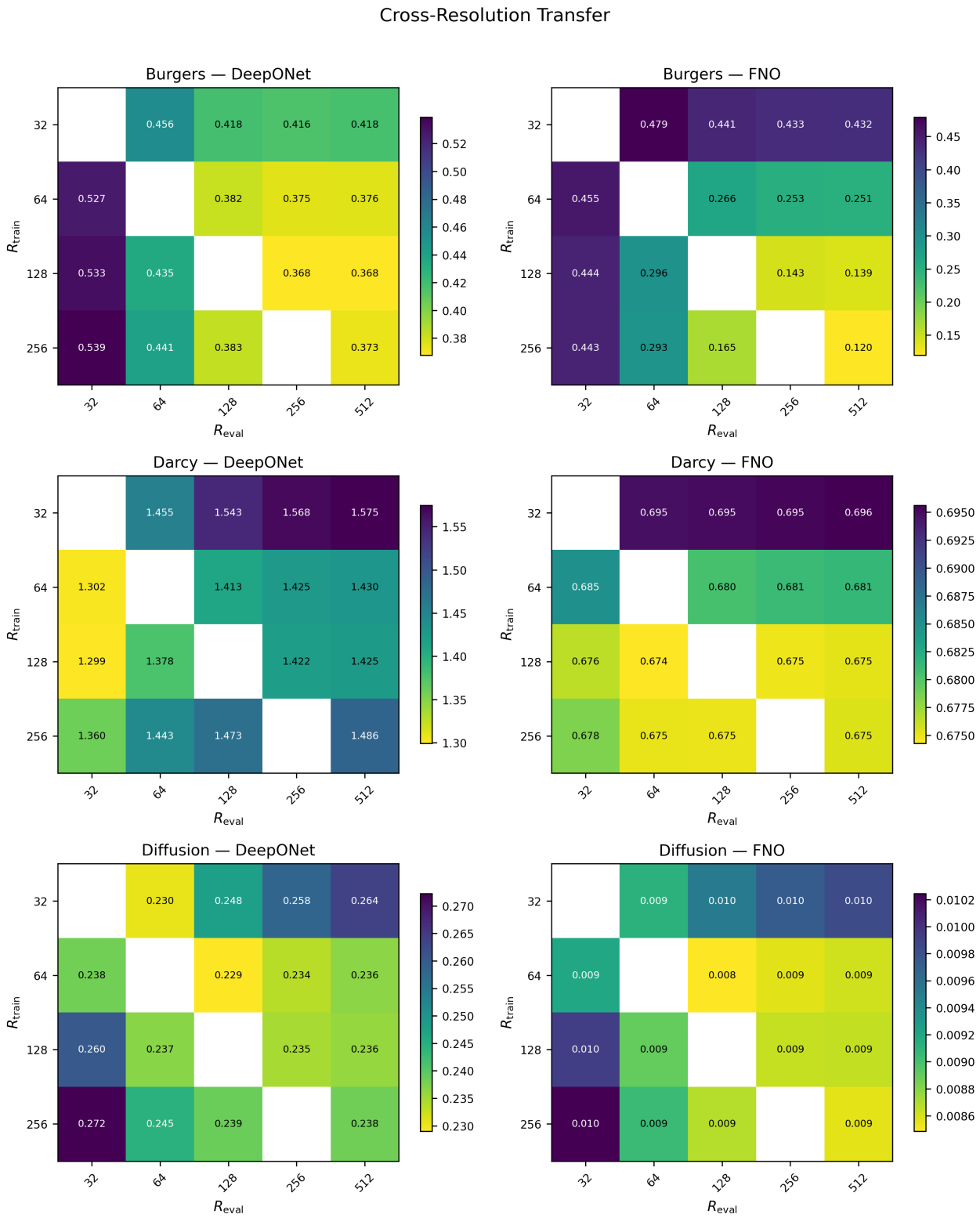

Figure 6: Cross-resolution transfer matrices. Each cell shows mean ε_{rel} for training at R_{train} (row) and evaluating at R_{eval} (column). Burgers–FNO shows strong asymmetric transfer (lower-right is best). Darcy and Diffusion are near-uniform.

Table 10: Held-out prediction contest: out-of-sample relative RMSE for power-law-only vs. AICc-best multi-law. Bold indicates lower OOS error.

Task	**Model**	**Axis**	**PL OOS**	**ML OOS**
Burgers	DeepONet	data	0.192	**0.192**
		capacity	**0.170**	0.170
		resolution	0.149	**0.148**
	FNO	data	0.305	**0.304**
		capacity	**0.338**	0.338
		resolution	0.267	**0.262**
	MLP	data	0.147	**0.147**
		capacity	0.247	**0.247**
		resolution	0.198	**0.192**
	MLP-ctrl	data	**0.382**	0.384
		capacity	**0.880**	0.880
		resolution	0.213	**0.195**
Darcy	DeepONet	data	0.137	**0.134**
		capacity	**0.303**	0.304
		resolution	**0.148**	0.149
	FNO	data	0.101	**0.097**
		capacity	0.264	0.264
		resolution	**0.327**	0.327
	MLP	data	0.113	**0.101**
		capacity	**0.236**	0.236
		resolution	0.358	**0.357**
Diffusion	DeepONet	data	0.172	0.172
		capacity	**0.211**	0.216
		resolution	0.322	**0.322**
	FNO	data	**0.118**	0.118
		capacity	0.362	0.362
		resolution	**0.321**	0.321
	MLP	data	**0.089**	0.089
		capacity	**0.428**	0.439
		resolution	**0.344**	0.344
	MLP-ctrl	data	**0.989**	0.990
		capacity	**2.859**	2.872
		resolution	0.224	**0.205**

Table 11: Within-slice bootstrap stability ($B = 500$): mean fraction of resamples selecting the dominant law. Values below 0.50 indicate the per-slice data is insufficient to distinguish law families.

Task	**Model**	**Data**	**Capacity**	**Resolution**
Burgers	DeepONet	0.35	0.43	**0.76**
	FNO	0.47	0.29	**0.52**
	MLP	0.43	0.37	**0.66**
Darcy	DeepONet	0.33	**0.54**	0.41
	FNO	0.41	**0.52**	0.45
	MLP	0.35	0.45	0.44
Diffusion	DeepONet	0.43	0.45	**0.51**
	FNO	0.45	0.42	0.45
	MLP	**0.59**	0.45	**0.59**

Table 12: Evidence-strength classification for every task–model–axis finding. "Robust": mean Akaike weight > 0.5 and within-slice bootstrap fraction > 0.5 (both thresholds met). "Suggestive": one threshold met. "Exploratory": neither met. Law identity is the AICc aggregate winner; parenthetical is the within-slice bootstrap mode when it disagrees.

Task	**Model**	**Axis**	**AICc Winner**	**Strength**
	DeepONet	data	exponential	Exploratory
	DeepONet	capacity	logarithmic (bs: linear)	Exploratory
	DeepONet	resolution	logarithmic	**Robust**
	FNO	data	power	Exploratory
Burgers	FNO	capacity	logarithmic	Exploratory
	FNO	resolution	linear (bs: log)	Suggestive
	MLP	data	saturation (bs: power)	Exploratory
	MLP	capacity	logarithmic	Suggestive
	MLP	resolution	logarithmic	**Robust**
	DeepONet	data	exponential (bs: log)	Exploratory
	DeepONet	capacity	logarithmic (bs: linear)	Suggestive
	DeepONet	resolution	linear (bs: log)	Exploratory
	FNO	data	stretched exp (bs: log)	Suggestive
Darcy	FNO	capacity	logarithmic	Suggestive
	FNO	resolution	logarithmic	Exploratory
	MLP	data	exponential (bs: power)	Suggestive
	MLP	capacity	logarithmic (bs: linear)	Exploratory
	MLP	resolution	logarithmic	Suggestive
	DeepONet	data	power (bs: log)	Exploratory
	DeepONet	capacity	linear	Exploratory
	DeepONet	resolution	logarithmic	Suggestive
	FNO	data	power	Exploratory
Diffusion	FNO	capacity	power (bs: linear)	Exploratory
	FNO	resolution	logarithmic	Exploratory
	MLP	data	power	**Robust**
	MLP	capacity	logarithmic (bs: linear)	Exploratory
	MLP	resolution	logarithmic	**Robust**

Robust: 4/27. Suggestive: 7/27. Exploratory: 16/27. The concentration of robust cases on the resolution axis (3/4) reflects its directional stability; data-axis and capacity-axis law identity is generally exploratory.

Table 13: R^2 for the additive 3D power law. High values are an artifact of the dominant data axis spanning most of the variance.

	MLP	**DeepONet**	**FNO**
Burgers	0.935	0.915	0.871
Darcy	0.886	0.788	0.880
Diffusion	0.951	0.842	0.893

Table 14: MLP and DeepONet capacity grid (hidden layers × hidden width). FNO uses a separate grid over modes, width, and number of layers.

Level	**MLP layers**	**Hidden width**	$\sim D$ **(MLP,** R=128**)**
tiny	2	32	∼5K
small	2	64	∼13K
small-med	2	96	∼22K
medium	2	128	∼34K
med-large	2	192	∼62K
large	2	256	∼82K
xlarge	3	256	∼148K

Table 15: AICc–BIC concordance rate: fraction of slices where both criteria select the same winning law, per axis.

Task	**Model**	**Data**	**Resolution**	**Capacity**
Burgers	MLP	4%	3%	90%
Burgers	DeepONet	21%	3%	85%
Burgers	FNO	11%	9%	70%
Darcy	MLP	4%	43%	80%
Darcy	DeepONet	11%	69%	95%
Darcy	FNO	14%	34%	65%
Diffusion	MLP	4%	97%	45%
Diffusion	DeepONet	11%	100%	80%
Diffusion	FNO	21%	74%	90%

The low concordance on the data axis ($n = 5$) and resolution axis ($n = 4$) reflects the small-n penalty inversion: at these sample sizes, BIC's complexity penalty is weaker than AICc's corrected penalty, so BIC systematically favors more complex models—the opposite of its large-n behavior. Capacity-axis concordance is generally high (>65%), consistent with the larger per-slice sample size ($n = 7$) where the penalty inversion is less severe. AICc remains the primary criterion throughout; BIC concordance is reported for transparency.

Table 16: Full (6-law) vs. restricted (3-law) winner on the resolution axis. "Direction" is the qualitative finding (helps / hurts / flat). Directional conclusions are robust to the candidate set.

Task	**Model**	**Full winner**	**Restricted winner**	**Direction robust?**
Burgers	MLP	logarithmic	logarithmic	✓
Burgers	DeepONet	logarithmic	logarithmic	✓
Burgers	FNO	linear	linear	✓
Darcy	MLP	logarithmic	logarithmic	✓
Darcy	DeepONet	linear	linear	✓
Darcy	FNO	logarithmic	logarithmic	✓
Diffusion	MLP	logarithmic	logarithmic	✓
Diffusion	DeepONet	logarithmic	logarithmic	✓
Diffusion	FNO	logarithmic	power	✓

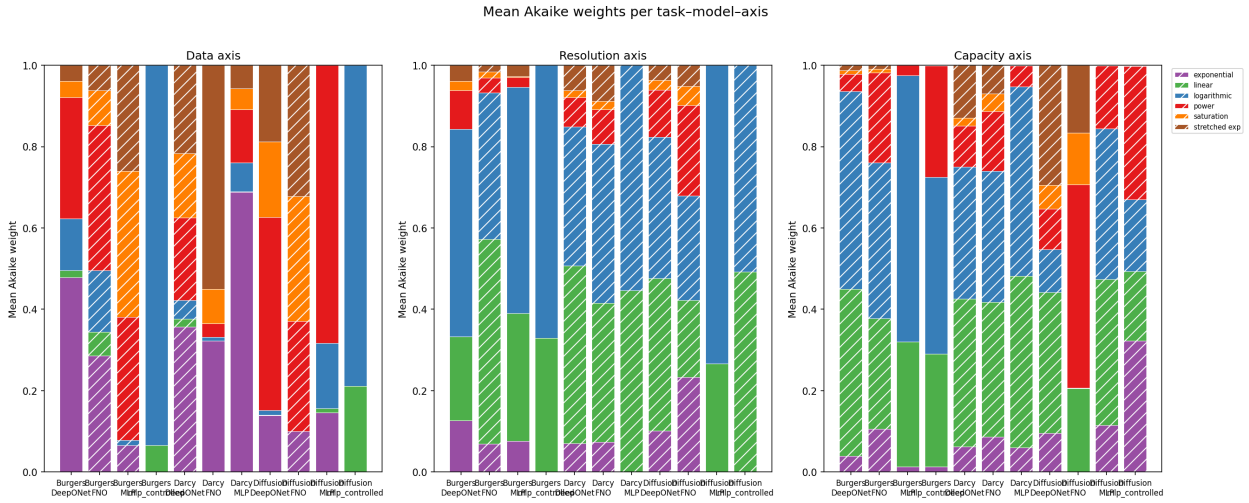

Figure 7: Mean Akaike weights across slices. Hatched bars indicate ambiguous cases where the top-two laws are within 15% of each other.

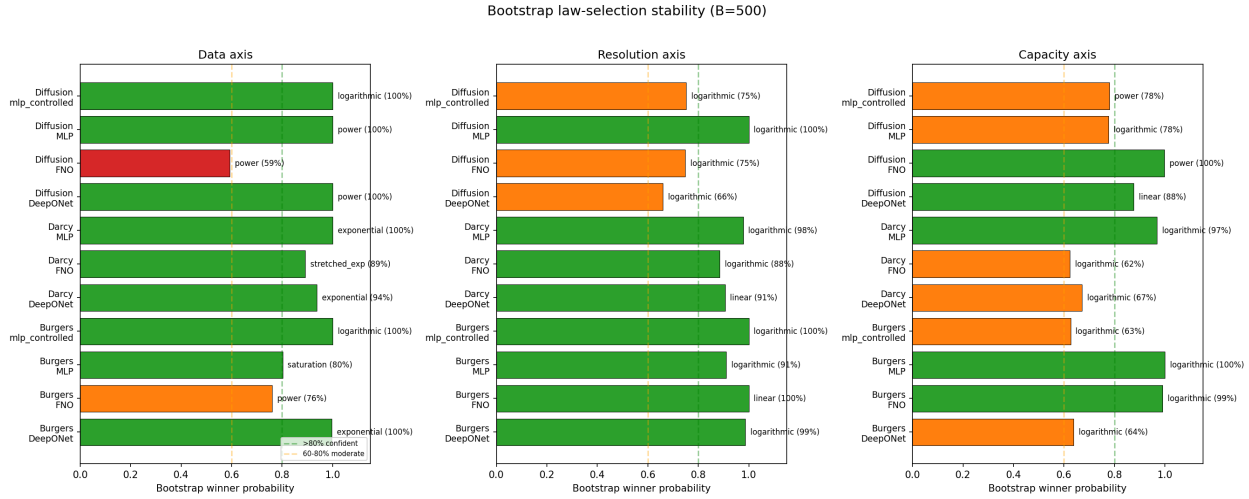

Figure 8: Bootstrap law-selection stability ($B = 500$). Green bars (>80%) indicate confident selection; orange (60–80%) moderate; red (<60%) ambiguous.

Table 17: Test relative L^2 error on Diffusion for the parameter-controlled MLP (`mlp_controlled`, top) vs. standard MLP baseline (bottom). Both show monotonically increasing error with R, ruling out the parameter-count confound.

Capacity	R=32	R=64	R=128	R=256
MLP Controlled (fixed parameter count)				
tiny	0.023	0.038	0.140	0.295
small	0.024	0.023	0.025	0.111
small-med	0.023	0.026	0.027	0.033
medium	0.021	0.027	0.027	0.026
med-large	0.020	0.025	0.033	0.031
large	0.018	0.024	0.031	0.034
xlarge	0.027	0.032	0.039	0.047
MLP Baseline (standard)				
tiny	0.118	0.118	0.125	0.132
small	0.096	0.101	0.105	0.103
small-med	0.088	0.094	0.098	0.100
medium	0.082	0.091	0.099	0.098
med-large	0.074	0.086	0.095	0.101
large	0.070	0.081	0.093	0.100
xlarge	0.102	0.117	0.127	0.133